

Group Project Assessment

COMP3020 Social Web Analytics

Total Marks: 25

Group Project (25%)

Topic Selection

Social media platforms like Reddit allow users to dive deep into niche discussions. To get started, pick a topic that interests you (e.g., Artificial Intelligence, Climate Change, Electric Vehicles, or Cryptocurrency).

It is recommended that each group selects a unique, active, and popular topic. This will facilitate a more comprehensive exploration of insights.

Advice: After downloading data from the required Social Web App, save the data to your working directory ('.csv' or '.json'). Otherwise, dynamic data changes will make your results irreproducible.

Text Mining

Question 1 (5 marks)

- a) **(1 mark)** Using the Reddit API, identify relevant thread URLs for your chosen topic. Focus on a monthly or yearly timeframe to ensure sufficient data depth.
- b) **(1 mark)** Find the top three threads with the highest Score (Upvotes). Retrieve and display the title and main post body of each of these three threads.
- c) **(2 marks)** Instead of individual words, context is key. Generate a Bigram (two-word sequence) frequency chart for the comments and replies within each of these threads. Display the top 15 most frequent bigrams for each thread.
- d) **(1 mark)** Comment on your Bigram charts. Do the frequent phrases indicate a positive, negative, or neutral sentiment in the discussion? Compare the tone between the three threads.

Clustering & Evaluation

Question 2 (8 marks)

- a) **(1 mark)** Combine all the comments from the top three threads collected in Question 1. Pre-process the comments (remove stopwords, punctuation, etc.) and construct a Document Term Matrix using TF-IDF weighting.
- b) **(2 marks)** According to your group, what is the optimal number of clusters for these comments? Use the Elbow Method analysis to justify your argument mathematically. Explain your reasoning in detail.

- c) **(1 mark)** Create a "Ground Truth" label for each comment based on its original thread.
- Label Thread 1 comments as "A"
 - Label Thread 2 comments as "B"
 - Label Thread 3 comments as "C"
- d) **(2 marks)** Apply K-means clustering to the combined dataset. You may need to use a dimension reduction method (PCA or MDS) before applying the K-means method to improve performance.
- e) **(1 mark)** Visualize the results of your clustering in a 2D vector space. Color the points by their assigned cluster but use different markers (shapes) to represent their original thread label.
- f) **(1 mark)** Comment on your findings. Did the K-means algorithm successfully reconstruct the original thread boundaries? Discuss any overlaps observed in the visualization.

Regression Analysis

Question 3 (4 marks)

Use all thread URLs on Reddit that you identified in Question 1 (not just the top 3).

- a) **(4 marks)** Test if there exists a linear relationship between the Score (Upvotes) of a thread and the Number of Comments it receives.
- Create a scatter plot to visualize this relationship.
 - Perform a linear regression analysis.
 - Interpret the R -squared value and the p-value. Based on your model, does a high Thread Score (Upvotes) guarantee high engagement in terms of comments?

Network Analysis (Mastodon)

Question 4 (6 marks)

- a) **(2 marks)** Using the Mastodon API, identify users related to your chosen topic. Identify the top three most popular users based on the highest number of Statuses (Posts) (i.e., the most active users, rather than just the most followed).
- b) **(2 marks)** Download 50 followers and 50 friends (users they follow) for these top 3 active users. Create a graph and visualize the network structure.

Note: Handle limitations such as hidden profiles or rate limits gracefully by documenting them.

- c) **(2 marks)** Calculate both the Degree Centrality and the PageRank for the nodes in your graph.
- Identify the top user for each metric.
 - Comment on the differences: Does the user with the most connections (Degree) also have the highest influence/quality connections (PageRank)?

Presentation & Submission

Presentation (2 marks)

Submission Requirements (2 marks for design and readability):

- **Cover Sheet:** Include a cover sheet with the Group Leader identified and all Student IDs listed.
- **Question Labeling:** Clearly label the question number (e.g., Question 2a) for each section.
- **Code Inclusion:** Include relevant snippets of code used for analysis. Do not dump the entire raw script; only include essential logic required to verify the output.
- **Output Presentation:** Display relevant outputs (charts, calculated metrics) directly in the report. Do not print raw datasets.
- **Comments:** Provide distinct "Answer/Conclusion" paragraphs interpreting your plots and hypothesis tests.